Heat Transfer

→ Heat is energy in transit due to temperature difference.

Unit = Joule ← 1 cal = 4.2 J

Dimension = $[ML^2T^{-2}]$ 1 kcal = 4200 J

* Spontaneously, heat will flow from higher temp to lower temp.

Ways for heat transfer:

① Convection: There is displacement of medium particles.

② Radiation:

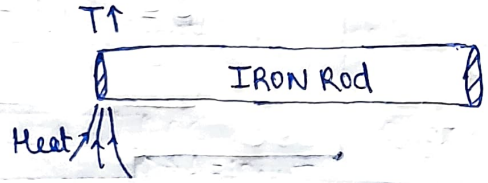


No medium is required in Radiation.

③ Conduction:

→ Medium is required.

→ No net displacement of medium particles.



When we heat one end, thermal agitations of molecules increases, ie they start vibrating faster and with greater amplitude. One molecule collides with other and makes it to vibrate as well. With this process, heat energy gets transferred from one end to other end. Temp. of other end also increases.

Thermal Current / Heat Current (i_{th}):



→ Unit = $\frac{\text{Joule}}{\text{sec}}$ = Watt

→ Dimension = $[ML^2T^{-3}]$

→ It has dir^n, in general from T_H to T_L

→ Scalar Quantity

Average thermal current:

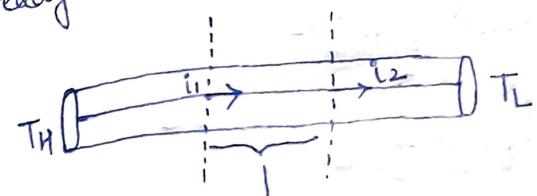
$t \to t+\Delta t \Rightarrow \Delta Q = \text{Heat} \Rightarrow i_{th} = \frac{\Delta Q}{\Delta t}$

Instantaneous thermal current:

$t \to t+dt \Rightarrow dQ = \text{Heat} \Rightarrow i_{th} = \frac{dQ}{dt}$

Steady State:

* In steady state $\boxed{i_1 = i_2}$



$$\frac{dQ}{dt} = i_1 - i_2 = 0$$

$$\Rightarrow \boxed{dQ = 0}$$

We know, $dQ = ms\,dT$

As $dQ = 0 \Rightarrow dT = 0$

$$\therefore \boxed{\frac{dT}{dt} = 0} \Rightarrow T = \text{constant}$$

But $T \neq$ uniform

In steady state:

1. Thermal current will be same in entire rod/wire.
2. Temperature will remain constant at a point ie at a point temp. will not change with time.
3. Temperature may not be uniform and changes with position.

Non-Steady State:



$i_1 \neq i_2$

① $i_1 > i_2$

$$\frac{dQ}{dt} = i_1 - i_2 > 0$$

$$\Rightarrow \boxed{dQ > 0}$$

Heat is absorbed

$dQ = ms\,dT$

If $dQ > 0 \Rightarrow dT > 0$

$$\therefore \frac{dT}{dt} = +ve \Rightarrow T \text{ increases with time}$$

② $i_1 < i_2$

$$\frac{dQ}{dt} = i_1 - i_2 < 0$$

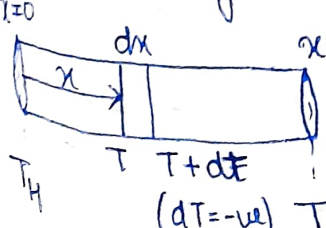
$$\Rightarrow \boxed{dQ < 0}$$

$$\Rightarrow \boxed{dT < 0}$$

$$\therefore \frac{dT}{dt} = -ve \Rightarrow T \text{ decreases with time}$$

Note: If it is given that temp is melting/boiling pt. then heat will used to change phase. $dQ = (dm)L$ (Latent heat = L)

Temperature gradient:



$$\boxed{\frac{dT}{dx} = \frac{T_L - T_H}{l} = -ve}$$

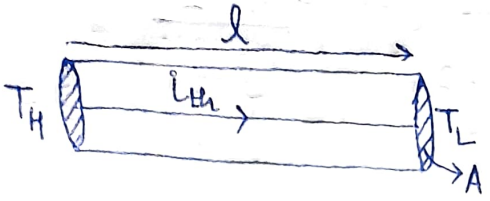
→ Unit = °C/m

→ Dimension = $[KL^{-1}]$

→ −ve along i_{th}

→ +ve opposite to i_{th}

Ohm's law:



$i_{th} \propto (T_H - T_L)$

$i_{th} \propto A$

$i_{th} \propto 1/l$

$$\therefore \boxed{i_{th} = \frac{dQ}{dt} = \frac{KA(T_H - T_L)}{l} = -KA\frac{dT}{dx}} \quad \left(\frac{dT}{dx} = \frac{T_L - T_H}{l}\right)$$

K = Thermal Conductivity

- → property of materials
- → For good conductor ⇒ K↑
- → For insulating material ⇒ K↓

→ Unit of $K = \frac{Watt}{m\,^\circ C} \equiv \frac{Watt}{m\text{-}Kelvin}$

→ Dimension = $[MLT^{-3}K^{-1}]$

Thermal Resistance (R_{th}):

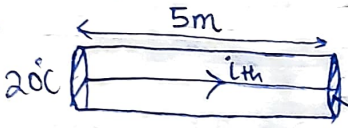
$$i_{th} = \frac{(T_H - T_L)}{(l/KA)} = \frac{T_H - T_L}{R_{th}} \Rightarrow \boxed{R_{th} = \frac{l}{KA}}$$

→ Unit = $^\circ C/watt = \frac{Kelvin}{watt}$

→ Dimension = $[M^{-1}L^{-2}T^{3}K]$

* R_{th} depends on material as well as shape, size, geometry of material

Q



Al rod, $K = \frac{400W}{m\text{-}K}$

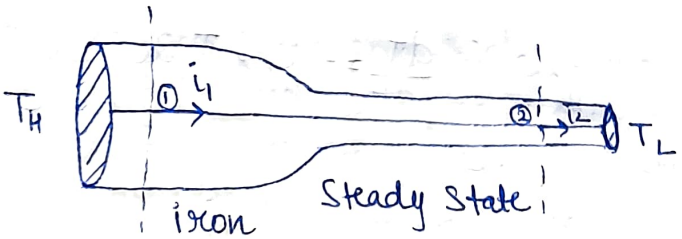
Find ① R_{th} ② i_{th}

① $R_{th} = \frac{l}{KA} = \frac{5m^2\text{-}K}{400W \times 10 \times 10^{-4}m^2}$

$R_{th} = \frac{50}{4}\frac{K}{watt} = 12.5\frac{Kelvin}{watt} = 12.5\frac{^\circ C}{watt}$

② $i_{th} = \frac{T_H - T_L}{R_{th}} = \frac{20^\circ c\ W}{12.5\ ^\circ} = \frac{8}{5}W$

Ex



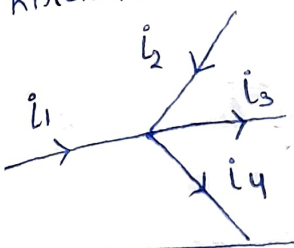
① $i_1 = i_2$ (steady state)

② $K_1 = K_2$ (same material)

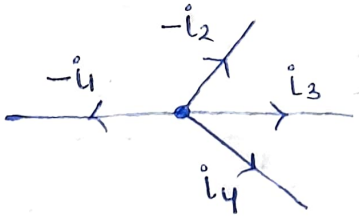
③ $\left|\frac{dT}{dx}\right|_① < \left|\frac{dT}{dx}\right|_②$ $\left(i_{th} = KA\left|\frac{dT}{dx}\right|\right.$ As $A \downarrow \Rightarrow \left|\frac{dT}{dx}\right| \uparrow$ to keep i_{th} constant$\left.\right)$

④ $\left(\frac{dT}{dt}\right)_① = \left(\frac{dT}{dt}\right)_② = 0$

Kirchoff current law (KCL):

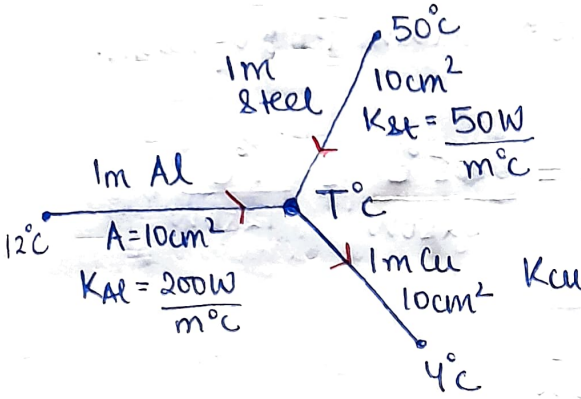


$$\sum i_{in} = \sum i_{out}$$
$$i_1 + i_2 = i_3 + i_4$$



$$\sum i_{out} = 0$$
$$i_3 + i_4 - i_1 - i_2 = 0$$

Q



Find temperature at junction (T).

At junction,

$$\sum i_{out} = 0$$
$$i_1 + i_2 + i_3 = 0$$

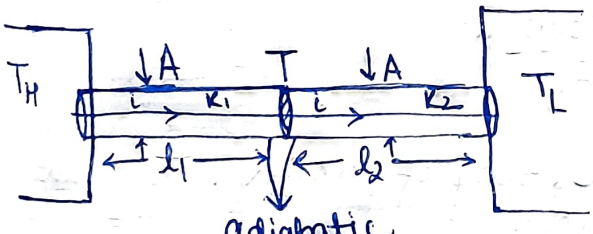
$$= \frac{(T-50)\times 50\times 10cm^2}{l} + \frac{(T-12)\times 200\times 10cm^2}{l} + \frac{(T-4\square)\times 400\times 10cm^2}{l} = 0$$

$$= T - 50 + 4T - 48 + 8T - 32 = 0$$

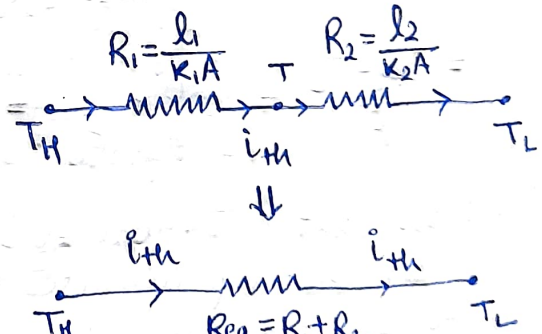
$$= 13T = 130 \Rightarrow \boxed{T = 10°C}$$

Combination of slabs:

① Series:



$\equiv$



$$i_{TH} = \frac{T_H - T_L}{R_{eq}} = \frac{T_H - T_L}{R_1 + R_2}$$

Now, $\frac{T_H - T_L}{R_1 + R_2} = \frac{T_H - T}{R_1}$

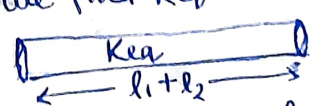
$$= T_H R_1 - T_L R_1 = T_H R_1 - T R_1 + T_H R_2 - T R_2$$

$$= T(R_1 + R_2) = T_H R_2 + T_L R_1$$

$$\boxed{T = \frac{T_H R_2 + T_L R_1}{R_1 + R_2}}$$

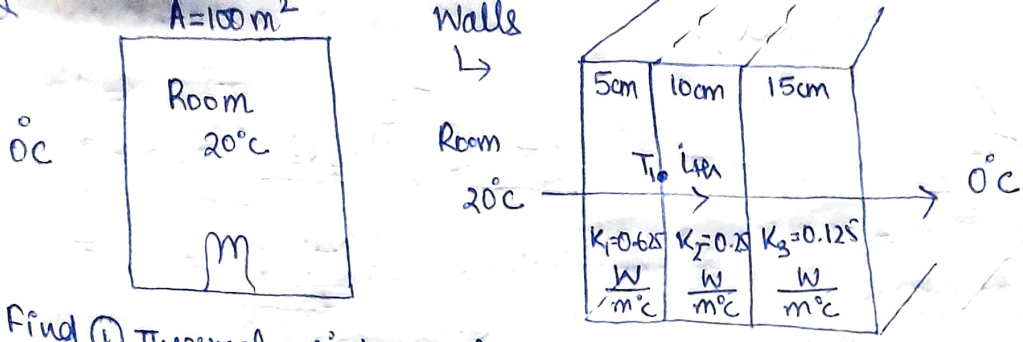
$$\boxed{R_{eq} = \frac{l_1}{K_1 A} + \frac{l_2}{K_2 A}}$$

Now we find K_{eq}:



$$\Rightarrow \frac{l_1 + l_2}{K_{eq} A} = \frac{l_1}{K_1 A} + \frac{l_2}{K_2 A}$$

$$\therefore \boxed{K_{eq} = \frac{l_1 + l_2}{\left(\frac{l_1}{K_1} + \frac{l_2}{K_2}\right)}}$$



Find ① Thermal resistance of each section of wall
② Net thermal resistance
③ Thermal current
④ Find power of heater required to maintain Temp of room.
⑤ Find T_1

①
$$R_1 = \frac{5\times10^{-2}\,m}{0.625\,\frac{W}{m^\circ C}\times 100\,m^2} = \frac{5\times10^{-2}\,^\circ C}{62.5\,W} = \frac{5^\circ C}{6250\,W} = \frac{1^\circ C}{1250\,W}$$

$$R_2 = \frac{10\times10^{-2}\,m}{0.25\,\frac{W}{m^\circ C}\times 100\,m^2} = \frac{0.1^\circ C}{25\,W} = \frac{1^\circ C}{250\,W}$$

$$R_3 = \frac{15\times10^{-2}\,m}{0.125\,\frac{W}{m^\circ C}\times 100\,m^2} = \frac{15\,^\circ C}{1250\,W} = \frac{3}{250}\,\frac{^\circ C}{W}$$

②
$$R_{eq} = R_1+R_2+R_3 = \frac{1}{1250}+\frac{1}{250}+\frac{3}{250} = \frac{21}{1250}\,\frac{^\circ C}{W}$$

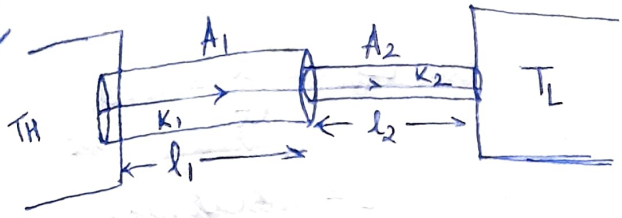
③
$$i_{th} = \frac{T_H - T_L}{R_{th}} = \frac{20^\circ C}{21/1250} = \frac{25000}{21}\,W$$

④ Now as heat goes from 20°C to 0°C, temp of room ↓. To maintain the room's temp we use a heater.
Power of heater must be same as the power we are losing as i_{th} so that the exact 20°C temp is maintained.

$$\boxed{P_{heater} = \frac{25000}{21}\,W}$$

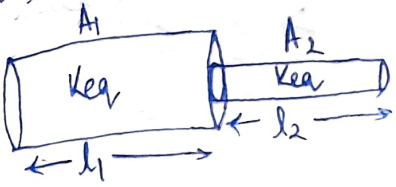
⑤
$$\frac{(20-T_1)}{1/1250} = \frac{25000}{21} \Rightarrow (20-T_1)1250 = \frac{25000}{21} \Rightarrow T_1 = \frac{400}{21}\,^\circ C$$

Ex



$$R_{eq} = R_1 + R_2 = \frac{l_1}{K_1 A_1} + \frac{l_2}{K_2 A_2}$$

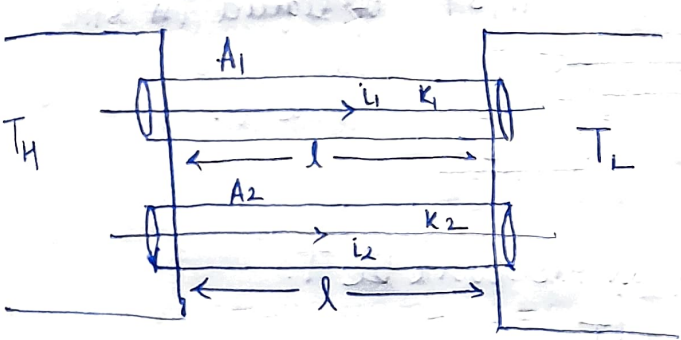
Now, K_{eq}:



$$\frac{l_1}{K_{eq} A_1} + \frac{l_2}{K_{eq} A_2} = \frac{l_1}{K_1 A_1} + \frac{l_2}{K_2 A_2}$$

$$K_{eq} = \frac{\frac{l_1}{A_1} + \frac{l_2}{A_2}}{\left(\frac{l_1}{K_1 A_1} + \frac{l_2}{K_2 A_2}\right)}$$

Parallel Combination:



$$i_1 = \frac{T_H - T_L}{R_1} \qquad i_2 = \frac{T_H - T_L}{R_2}$$

$$i = i_1 + i_2 = (T_H - T_L)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

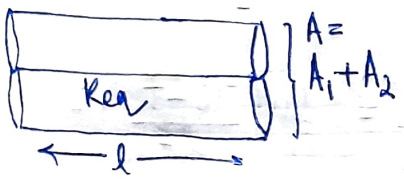
$$\therefore (T_H - T_L)\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{T_H - T_L}{R_{eq}}$$

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{i_1}{i_2} = \frac{R_2}{R_1}$$

$$\frac{1}{R_{eq}} = \frac{K_1 A_1}{l} + \frac{K_2 A_2}{l}$$

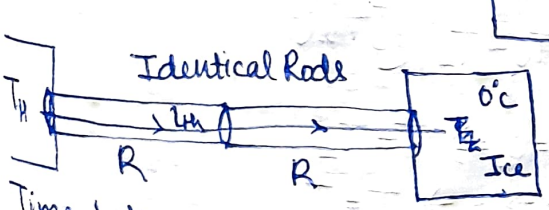
Now K_{eq}:



$$\frac{K_{eq}(A_1 + A_2)}{l} = \frac{K_1 A_1 + K_2 A_2}{l}$$

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$$

Q



Time taken to melt ice = 10 mins

$$i_{TH} = \frac{T_H - 0}{R + R} = \frac{T_H}{2R} \Rightarrow \Delta Q = i_{TH} \times t = \frac{T_H}{2R} \times 10 \times 60 = mL_f$$

(series)

$$i'_{TH} = \frac{T_H - 0}{1/\left(\frac{1}{R} + \frac{1}{R}\right)} = \frac{T_H}{R/2} = 4 i_{TH} \Rightarrow \Delta Q' = \frac{T_H}{R} \times 2 \times t = mL_f$$

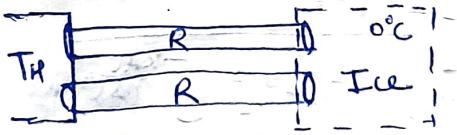
(parallel)

Now $\Delta Q = \Delta Q'$

$$\frac{T_H}{2R} \times 10 \times 60 = \frac{T_H}{R} \times 2 \times t$$

$$t = \frac{10 \times 60}{2 \times 2}$$

$$t = 2.5 \text{ mins}$$



Find time taken to melt ice.

Solids/liquids

$dQ = ms\,dT$

$\boxed{dQ = c\,dT}$ (c = heat capacity)

$i_{th} = \frac{dQ}{dt} = ms\frac{dT}{dt}$ ($\frac{dT}{dt}$ = Rate of change of temp.)

At melting point/boiling point:

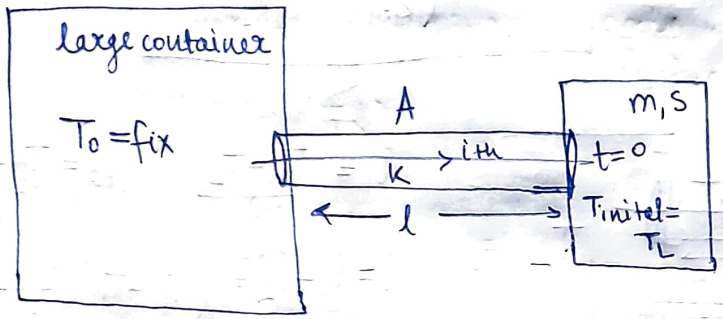
$i_{th} = \frac{dQ}{dt} = \frac{dm}{dt}L$ (L = Latent heat)

For gases:

$i_{th} = \frac{dQ}{dt} = nc\frac{dT}{dt}$

c = molar heat capacity
↳ process dependent.

* Q. $T_0 > T_1$



Find temperature of liquid in small reservoir at any time t.

Let at any time t, temp of small reservoir be T.

$i_{th} = \frac{(T_0 - T)KA}{l} = \frac{dQ}{dt} = ms\frac{dT}{dt}$

$\Rightarrow \frac{KA}{msl}\int_0^t dt = \int_{T_L}^{T}\frac{dT}{T_0 - T} \Rightarrow \frac{KAt}{msl} = -\ln(T_0 - T)\Big|_{T_L}^{T}$

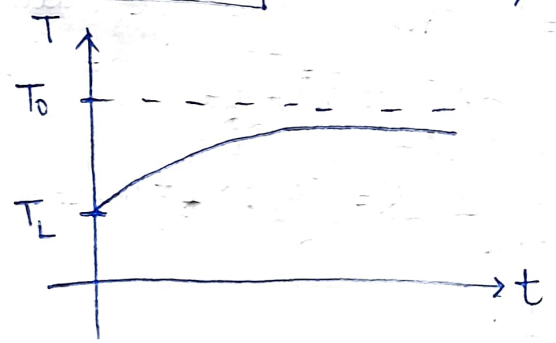
$= -\frac{KAt}{msl} = \ln(T_0 - T) - \ln(T_0 - T_L) \Rightarrow -\frac{KAt}{msl} = \ln\left(\frac{T_0 - T}{T_0 - T_L}\right)$

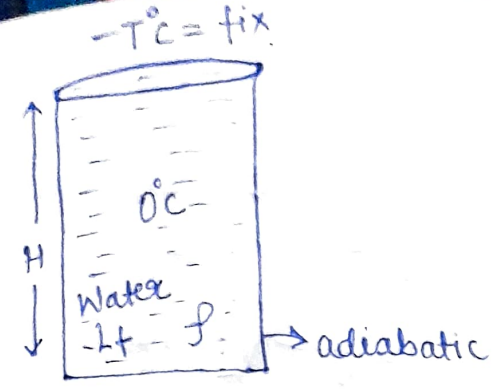
$= \frac{T_0 - T}{T_0 - T_L} = e^{\frac{-KAt}{msl}} \Rightarrow T_0 - T = (T_0 - T_L)e^{\frac{-KAt}{msl}}$

$\boxed{T = T_0 - (T_0 - T_L)e^{\frac{-KAt}{msl}}}$

At $t = 0 \Rightarrow T = T_L$

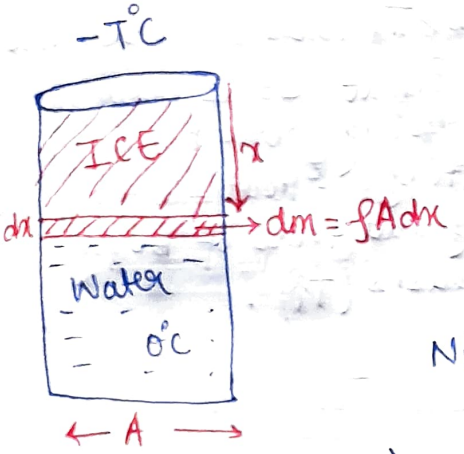
At $t \to \infty \Rightarrow T = T_0$





Given, thermal conductivity of ice = K
Find time taken by the water to completely freeze.

Assumption: Here we assume the density of ice to be just slightly less than water, so the freezing of water will start from the top of container.
For calculation ρ density of ice = ρ



Ice acts as resistance here



$$i_{th} = \frac{0-(-T)}{x} KA = \frac{KAT}{x}$$

Now, $i_{th} = \frac{KAT}{x} = \frac{dQ}{dt} = \frac{dm}{dt} L_f$

$$\Rightarrow \frac{KAT}{x} = \frac{\rho A dx}{dt} L_f \Rightarrow \frac{KT}{x} = \frac{\rho dx}{dt} L_f$$

$$\Rightarrow \frac{KT}{\rho L_f} \int_0^t dt = \int_0^H x\,dx \Rightarrow \frac{KT}{\rho L_f} t = \frac{H^2}{2} \Rightarrow \boxed{t = \frac{\rho L_f H^2}{2KT}}$$

Radiations:
→ No medium is required
→ Intervening medium is not heated
→ Radiations are electromagnetic waves.
→ Speed of radiation in vacuum is 'c' and in any other medium is c/n. (n = refractive index)
→ Light is part of radiation.
→ Any object having temperature more than 0Kelvin will emit radiations.

Maxwells theory suggests the reason for emission of EM wave/radiations is vibration of charge particles in an atom or molecule.

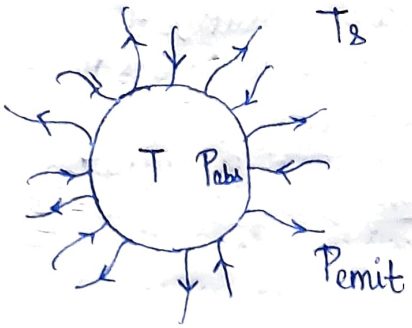
$$T\uparrow \Rightarrow KE\uparrow \Rightarrow \text{Vibration} \Rightarrow \text{Radiations}\uparrow$$

Also $E = \frac{hc}{\lambda} \rightarrow E\uparrow, \lambda\downarrow \Rightarrow \boxed{T\uparrow \Rightarrow \lambda\downarrow}$

→ At room temp: Radiations are in infrared region
For luminious objects: Radiations are in light region
For non-luminious objects: Radiations are in region after IR and before UV.

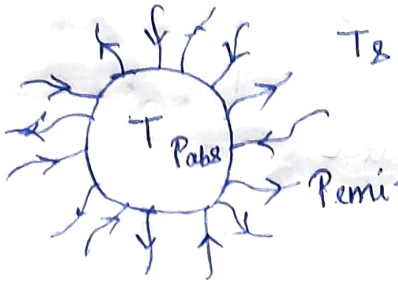
Prevost Theory of exchange:

① $T > T_s$



As $T > T_s$, P_{emit} will be more than P_{abs}. So as body is emitting more power than absorbing, heat flows out of it and temp. of body decreases ($T\downarrow$).

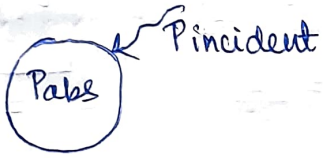
② $T_s > T$



As $T_s > T$, P_{abs} will be more than P_{emit}. So as body is absorbing more power than emitting, & heat flows in the body and temp of body increases ($T\uparrow$).

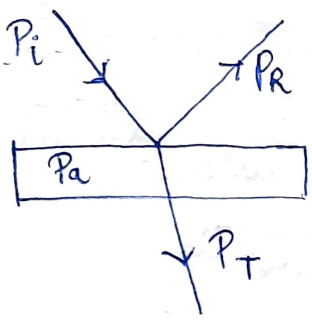
Perfectly Black Body:

→ P.B.B. allows all the radiations incident on it.
→ P.B.B. absorbs all radiations, reflect or transmit none.
→ P.B.B. also emits radiations.



$$\boxed{P_{abs} = P_{incident}}$$

Absorptive Power:



By E/C ⇒ $P_i = P_a + P_R + P_T$

$$1 = \frac{P_a}{P_i} + \frac{P_R}{P_i} + \frac{P_T}{P_i}$$

$$\therefore \boxed{a + r + t = 1}$$

$$\begin{cases} a = \frac{P_a}{P_i} \\ r = \frac{P_R}{P_i} \\ t = \frac{P_T}{P_i} \end{cases}$$

a = Absorptive Power

$$a = \frac{P_a}{P_i} \quad (0 \le a \le 1)$$

- → Unitless
- → Dimensionless
- → Ratio

For PBB ⇒ $P_a = P_{in}$

$$\therefore \boxed{a = 1}$$

$$\boxed{r = t = 0}$$

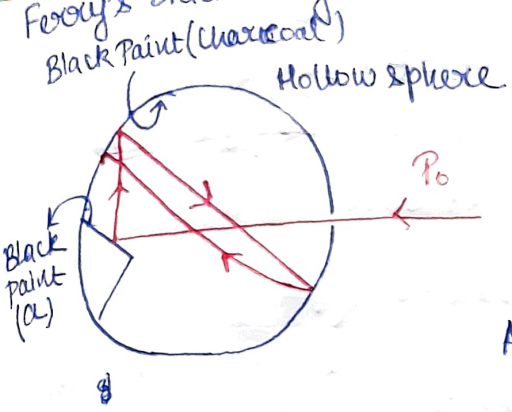
Purely reflecting:

$P_R = P_i \Rightarrow r = 1, a = 0, t = 0$

Purely transparent:

$P_T = P_i \Rightarrow t = 1, a = 0, r = 0$

Fevoy's Black Body:



Let radition make n-collisions.

After first collision, $P_{abs} = aP_0$

∴ Power of radiation becomes $= (1-a)P_0$

After second collision, power of radiation $= (1-a)^2 P_0$

After n-collisions, power of radiation $= (1-a)^n P_0$

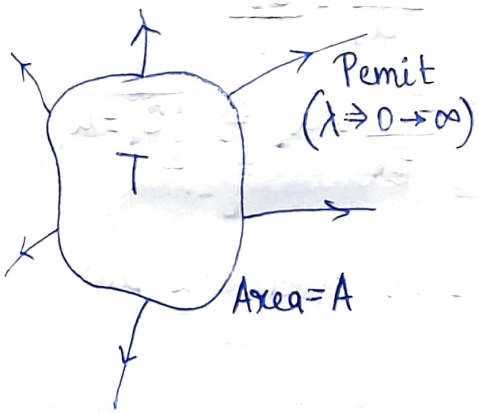
$P_{abs} = P_0 - (1-a)^n P_0$

$$a_{effective} = \frac{P_{abs}}{P_0} = 1-(1-a)^n$$

Now, for ex. let $a = 0.8$ and $n = 10$

$a_{eff} = 1-(0.2)^n = 1-\left(\frac{1}{5}\right)^{10} = 1-\frac{1}{5^{10}} \approx 1$ (very small)

Emissive Power (E):



* $T\uparrow \Rightarrow$ no. of photons with low $\lambda\uparrow$
  $T\downarrow \Rightarrow$ no. of photons with high $\lambda\uparrow$

$$E = \frac{P_{emit}}{A} = \text{Emissive Power}$$

→ Unit = Watt/m^2

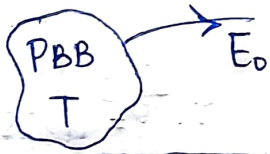
→ Dimension = $[MT^{-3}]$

Note:

* $E \propto T^4$ (T in Kelvin)

Emissivity (e):

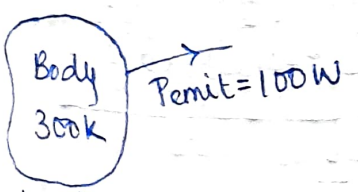


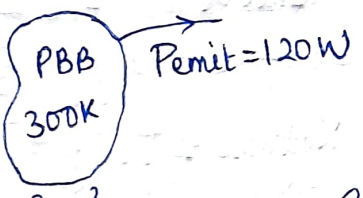


$$e\{\text{of body}\} = \frac{E}{E_0} = \frac{\text{emissive power of body at temp T.}}{\text{emissive power of PBB at temp T.}} \quad (0 \le e \le 1)$$

→ Unitless, dimensionless

Ex



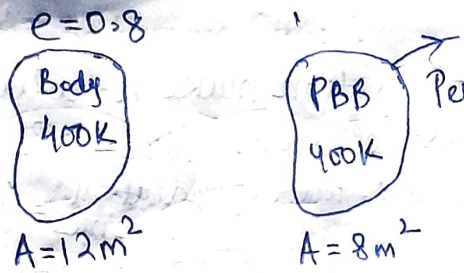


Find e.

$$e = \frac{E}{E_0} = \frac{100/10}{120/8}$$

$$e = \frac{8}{12} = \frac{2}{3} = 0.67$$

Ex



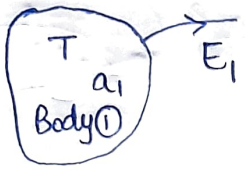
Find Pemit by body.

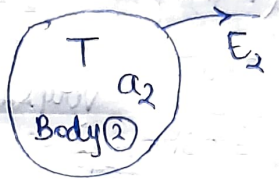
$e = 0.8 = \frac{P/12}{50/8} = \frac{P \times 8}{50 \times 12}$

$\boxed{P = 60\,W}$

Kirchoff's Law:

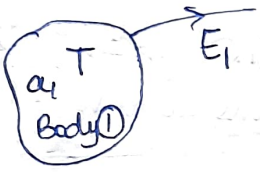
→ It states that any two bodies at same temp. will have equal ratio of emissive power to absorptive power.

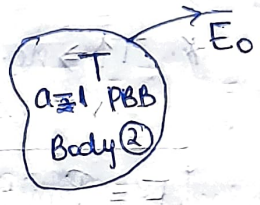




$\star \boxed{\frac{E_1}{a_1} = \frac{E_2}{a_2}}$

Now,





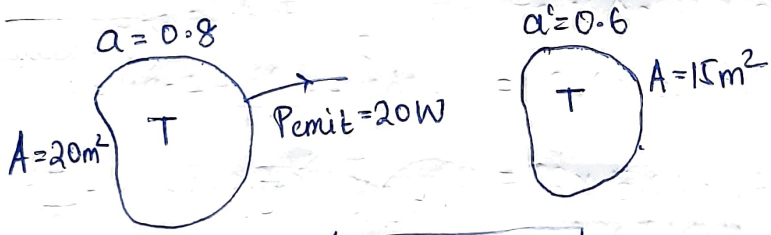
$\frac{E_1}{a_1} = \frac{E_0}{1}$

$\frac{E_1}{E_0} = a_1 \Rightarrow \boxed{e_1 = a_1}$

$\therefore \star \boxed{a = e}$

→ Good absorbers are good emitters and bad absorbers are bad emitters.

Q



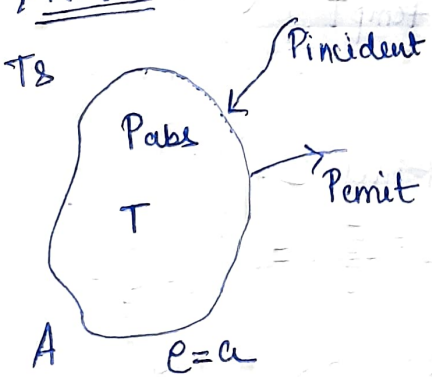
$\therefore \boxed{P = 11.25\,W}$

Find Pemit by 2nd body. Ex

$\frac{E_1}{a_1} = \frac{E_2}{a_2}$

$\frac{20/20}{0.8} = \frac{P/15}{0.6}$

★Note:

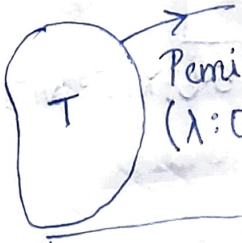


Pemit depends on:
1. Temp of body
   $P_{emit} \propto T^4$
2. Surface area (A)
3. Nature of body
   $e \uparrow \Rightarrow P_{emit} \uparrow$

Pincident depends on:
1. Temp. of surrounding
   $P_{incident} \propto T_s^4$
2. Surface Area (A)

Pabs depends on:
1. T_s
2. Surface area A
3. Nature of body (a)

Spectral Emissive Power (E_λ):



$$E = \frac{P_{emit}}{A}$$

Now, we take a small wavelength:

$\lambda \to \lambda + d\lambda \Rightarrow dE$

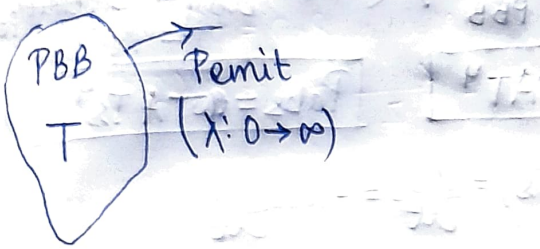
$$\star \quad \boxed{E_\lambda = \frac{dE}{d\lambda}}$$

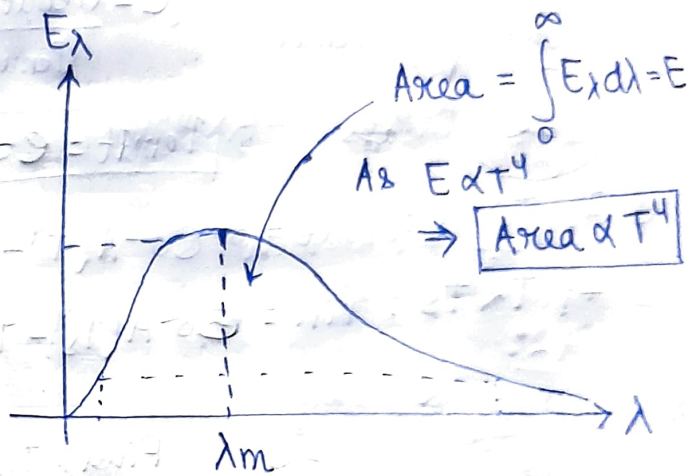
Also, $$\boxed{E = \int dE = \int_0^\infty E_\lambda \, d\lambda}$$

Area under E_λ v/s λ curve gives E.

Also, Area $\propto T^4$

Wien's Displacement Law:





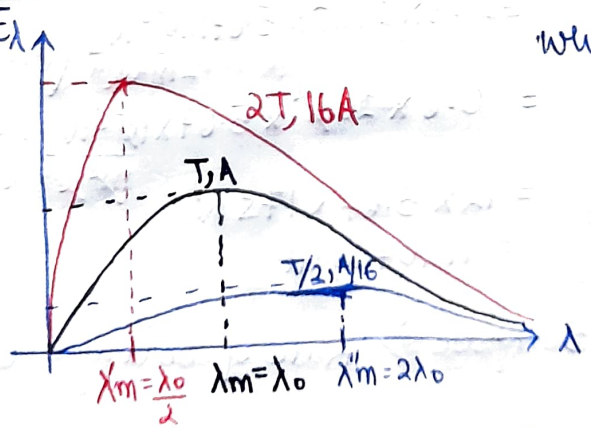
$$\boxed{\lambda_m \propto \frac{1}{T(\text{kelvin})}} \quad (T\uparrow, \lambda\downarrow)$$

λ_m → Wavelength corresponding to max^m emission.

$$\therefore \quad \boxed{\lambda_m T = b}$$

b = Wein's Constant = $0.282\,cm\text{-}K = 2.82\times10^{-3}\,m\text{-}K$

Ex.



When T becomes 2T

$\lambda'_m = \frac{\lambda_0}{2}$ as $\lambda_m \propto \frac{1}{T}$

Also, Area $\propto T^4$

Area $\to 2^4 A = 16A$

$E \to 16E$

Similarly for $T \to T/2$.

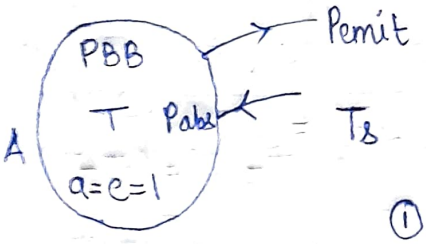
Q. If radiations emitted by sun are max^m at 4700 Å, then find temp. of sun.

$\lambda_m T = b$

$4700\times10^{-10}\,m \; T = 2.82\times10^{-3}\,m\text{-}K$

$$T = \frac{2.82\times10^{-3}}{4700\times10^{-10}}\,K = \frac{2.82\times10^{5}}{47}\,K = \frac{282}{47}\times1000\,K = 6000\,\text{Kelvin}$$

Stefan-Boltzman law :



$$P_{emit} = EA = \sigma A T^4$$

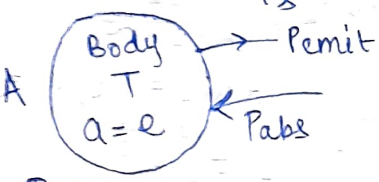
σ = Stefan's Const = $5.67 \times 10^{-8}\ W/m^2K^4$

$$P_{abs} = \sigma A T_s^4$$

① If $T > T_s$: $P_{out} = \sigma A(T^4 - T_s^4) = \frac{dQ}{dt} = -ms\frac{dT}{dt}$

② If $T < T_s$: $P_{in} = \sigma A(T_s^4 - T^4) = \frac{dQ}{dt} = ms\frac{dT}{dt}$

Now,

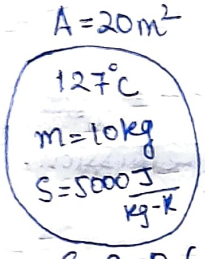


$$e = \frac{(P_{emit})_{body}}{(P_{emit})_{PBB}} \Rightarrow (P_{emit})_{body} = e(P_{emit})_{PBB}$$

$$P_{emit} = e\sigma AT^4 \qquad P_{abs} = e\sigma A T_s^4$$

① If $T > T_s$: $P_{out} = e\sigma A(T^4 - T_s^4) = \frac{dQ}{dt} = -ms\frac{dT}{dt}$

② If $T < T_s$: $P_{in} = e\sigma A(T_s^4 - T^4) = \frac{dQ}{dt} = ms\frac{dT}{dt}$

Q. 

27°C

Find ① The net power outgoing through radiation
② Rate of cooling $\left(\frac{dT}{dt}\right)$.

① $P_{out} = eA\sigma T^4 - eA\sigma T_s^4 = eA\sigma(T^4 - T_s^4)$

$= 0.6 \times 20m^2 \times 5.67\times10^{-8}\ \frac{W}{m^2K^4}\left((400)^4 - (300)^4\right)K^4$

$= 0.6 \times 20m^2 \times 5.67\times10^{-8}\ \frac{W}{m^2K^4}(256 - 81)K^4 \times 10^8$

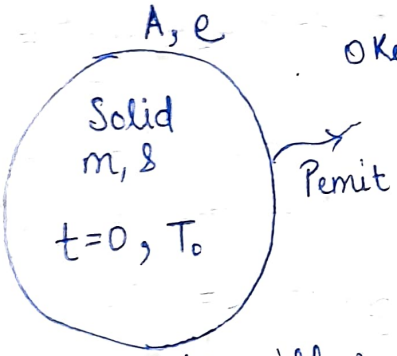
$= 12 \times 5.67 \times 175\ W$

$= 11907$ Watt

② $P_{out} = 11907 = \frac{dQ}{dt} = -ms\frac{dT}{dt}$

$11907\ \frac{J}{s} = -10kg \times \frac{5000J}{Kg-K} \times \frac{dT}{dt}$

$\frac{dT}{dt} = \frac{-11907}{50000}\ \frac{K}{sec} = -0.238\ K/sec \Rightarrow$ Rate of cooling $= 0.238\ K/sec$

Q 

0 Kelvin — Find temp of solid at any time t.

(As $T_s = 0K$, it will only emit radiations)

$$P_{emit} = \sigma e A T^4 = \frac{dQ}{dt} = -ms\frac{dT}{dt}$$

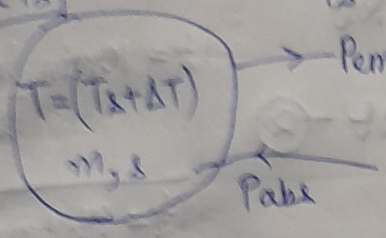
$$= -\frac{\sigma eA}{ms}\int_0^t dt = \int_{T_0}^{T} T^{-4}dT$$

$$\frac{e\sigma A_0 t}{ms} = \frac{(T^{-3})_{T_0}^{T}}{3} \Rightarrow \frac{3e\sigma A_0 t}{ms} = \frac{1}{T^3} - \frac{1}{T_0^3}$$

$$\boxed{\frac{1}{T^3} = \frac{1}{T_0^3} + \frac{3e\sigma A_0 t}{ms}}$$

Special case of Stefan's-Boltzmann law ⇒ Newtons law of cooling

$\Delta T \ll T_s$



$P_{emit} = e\sigma A (T_s+\Delta T)^4$

$P_{abs} = \sigma e A T_s^4$

$P_{out} = e\sigma A \{(T_s+\Delta T)^4 - T_s^4\}$

$= e\sigma A T_s^4 \{(1+\frac{\Delta T}{T_s})^4 - 1\}$

using $(1+x)^n \approx (1+nx)$

$= e\sigma A T_s^4 (1 + \frac{4\Delta T}{T_s} - 1)$

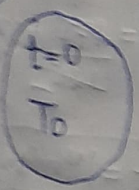
$\therefore \boxed{P_{out} = 4e\sigma A T_s^3 \Delta T}$

Now, $P_{out} = \frac{dQ}{dt} = -ms\frac{dT}{dt}$

$4e\sigma A T_s^3 \Delta T = -ms\frac{dT}{dt}$

$$\boxed{\frac{dT}{dt} = -\frac{4e\sigma A T_s^3}{ms}\Delta T}$$

$$\therefore \boxed{\text{Rate of cooling} = \frac{dT}{dt} = -K\Delta T = -K(T-T_s)}$$



Assuming newtons law of cooling to be valid, find temp. of body at any time t.

Let at any time t, temp of body be T.

$$\frac{dT}{dt} = -K(T-T_s) \Rightarrow \int_{T_0}^{T} \frac{dT}{T-T_s} = -K\int_0^t dt$$

$= \ln(T-T_s)_{T_0}^{T} = -K(t)_0^t$

$= \ln(T-T_s) - \ln(T-T_0) = -Kt$

$= \ln\left(\frac{T-T_s}{T-T_0}\right) = -kt$

$= \frac{T-T_s}{T-T_0} = e^{-kt}$

$\therefore \boxed{T = T_s + (T_0 - T_s)e^{-kt}}$

$\therefore \boxed{(T-T_s) = (T_0-T_s)e^{-kt}}$

Q

40°C

$T_s = 20°C$

Temp. of object decreases from 40°C to 35°C in 10mins. Find time taken for temp to decrease from 35°C to 30°C.

Method 1:

Using NLC:

$$(T - T_s) = (T_0 - T_s)e^{-kt}$$

$$\ln\left(\frac{T - T_s}{T_0 - T_s}\right) = -kt$$

For 40°C to 35°C →

$$\ln\left(\frac{35-20}{40-20}\right) = -k \times 10\,min$$

$$\ln\left(\frac{15}{20}\right) = \ln\left(\frac{3}{4}\right) = -k \times 10\,min$$

$$\therefore \ln\left(\frac{4}{3}\right) = k \times 10\,min \quad -(1)$$

For 35°C to 30°C

$$\ln\left(\frac{30-20}{35-20}\right) = -kt$$

$$\ln\left(\frac{15}{10}\right) = kt$$

$$\ln\left(\frac{3}{2}\right) = kt \quad -(2)$$

Dividing (2) by (1)

$$\frac{kt}{k \times 10\,min} = \frac{\ln(3/2)}{\ln(4/3)}$$

$$t = 10\,min\,\frac{\ln(3/2)}{\ln(4/3)}$$

$$\boxed{t = 14.4\,mins}$$

Method 2:

Using approximation

NLC: $\frac{dT}{dt} = -k(T - T_s)$

For 40°C to 35°C

$$\frac{-5°C}{10\,min} = -k\left(\frac{40+35}{2} - 20\right)°C$$

(avg)

$$\frac{5°C}{10\,min} = k \times 17.5 \quad -(1)$$

For 35°C to 30°C

$$\frac{-5°C}{\Delta t} = -k\left(\frac{35+30}{2} - 20\right)$$

$$\frac{5°C}{\Delta t} = k \times 12.5 \quad -(2)$$

Dividing (1) and (2)

$$\frac{5 \times \Delta t}{10\,min \times 5} = \frac{17.5}{12.5}$$

$$\Delta t = \frac{175}{125} \times 10$$

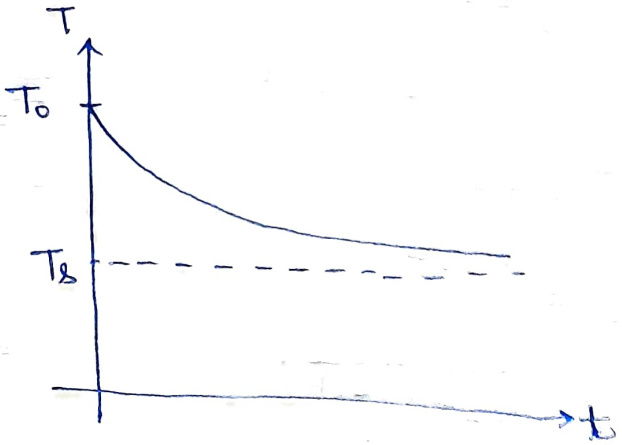
$$\boxed{\Delta t = 14\,mins}$$

Graph of Temperature variation:

$$\boxed{T = T_s + (T_0 - T_s)e^{-kt}}$$

At $t = 0$ $T = T_0$

At $t \to \infty$ $T = T_s$



Solar Constant:

→ We take sun as a perfectly black body.

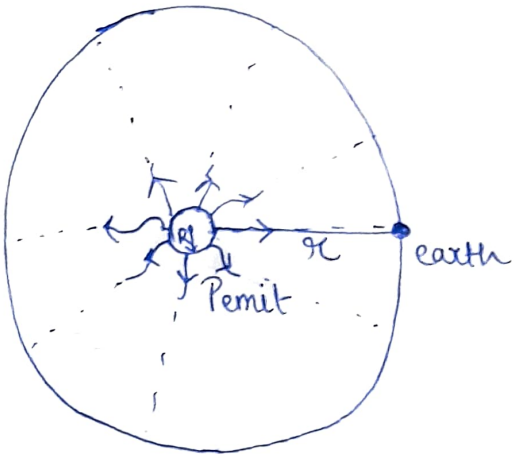
→ Radius of Sun = R

→ Temp. of Sun = T

→ Pemit = $e\sigma A T^4$ $(e=1)$

$= \sigma 4\pi R^2 T^4$

→ x = sep b/w sun and earth



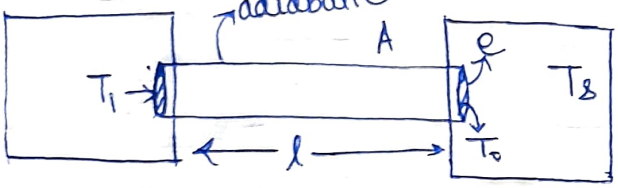
Intensity of Radiation on earth $= \frac{\text{Pemit}}{4\pi x^2} = \frac{\sigma 4\pi R^2 T^4}{4\pi x^2}$

$\therefore$ $\boxed{I = \sigma \frac{R^2}{x^2} T^4 = S = \text{Solar Constant}}$

↳ Power incident on earths surface per unit cross sectional area.

$\star$ $\boxed{S = 1400 \frac{\text{Watt}}{m^2}}$

Q



System is at steady state.
Find thermal conductivity of rod.

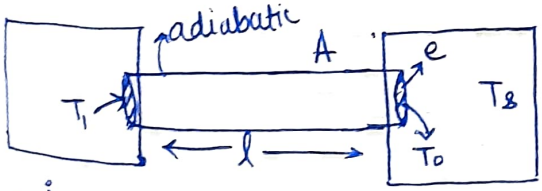
$i_{conduction} = \frac{(T_1 - T_0) KA}{\ell}$

$i_{radiation} = \frac{dQ}{dt} = eA\sigma (T_0^4 - T_s^4)$

System is at steady state $\Rightarrow$ $i_{conduction} = i_{radiation}$

$\frac{(T_1 - T_0) KA}{\ell} = eA\sigma (T_0^4 - T_s^4) \Rightarrow \boxed{K = e\sigma \ell \frac{(T_0^4 - T_s^4)}{(T_1 - T_0)}}$

Q



System is at steady state.
NLC is valid $(T_0 = T_s + \Delta T)$
Find thermal conductivity of rod.

$i_c = i_r$

$= \frac{KA(T_1 - T_0)}{\ell} = eA\sigma (T_0^4 - T_s^4) = eA\sigma \left((T_s + \Delta T)^4 - T_s^4\right) = eA\sigma \left(T_s^4 \left\{\left(1 + \frac{\Delta T}{T_s}\right)^4 - 1\right\}\right)$

$\Rightarrow \frac{KA(T_1 - T_0)}{\ell} = eA\sigma T_s^4 \left(1 + \frac{4\Delta T}{T_s} - 1\right) = eA\sigma T_s^3 4\Delta T$

$\Rightarrow \frac{KA(T_1 - T_0)}{\ell} = 4e\sigma A T_s^3 \Delta T$

$\boxed{K = \frac{4e\sigma T_s^3 \Delta T \ell}{T_1 - T_0}}$